

# Backfire of Influence Manipulation: How Manipulating Moderate Influencers Affects Online Discourse

Tomoya Takeda<sup>1</sup>, Fujio Toriumi<sup>2</sup>, Toshiharu Sugawara<sup>1</sup>

<sup>1</sup> Department of Computer Science and Communications Engineering,  
Waseda University, 3-4-1 Okubo, Shinjyuku-ku, 169-8555, Tokyo, Japan.

`tomoya.leo9@fuji.waseda.jp, sugawara@waseda.jp`

<sup>2</sup> Department of Systems Innovation, The University of Tokyo,  
7-3-1 Hongo, Bunkyo-ku, 113-8656, Tokyo, Japan.

`tori@sys.t.u-tokyo.ac.jp`

**Abstract.** We investigate the effect of externally manipulated hub users (influencers) on public opinion in online social media, and discuss how this triggers polarization through repulsion from opposing views. By integrating established models of opinion dynamics (Friedkin-Johnsen, bounded confidence, and social reinforcement), we reveal an emergent backfire effect—a counter-intuitive polarization mechanism. Experiments on scale-free networks show that steering influencers with moderate opinions toward extremes provokes repulsive responses from users with opposing views, depending on the influencers’ network position. These findings highlight how targeted manipulation of well-connected, neutrally positioned influencers can unintentionally intensify polarization, highlighting the structural role of influencers in shaping opinion polarization.

**Keywords:** Social Networks, Scale-free Networks, Multi-Agent Systems, Computational Social Science, Opinion Dynamics, Polarization

## 1 Introduction

Social media, such as *consumer-generated media*, has become pivotal in shaping individuals’ decision-making processes. Social media platforms such as Twitter (currently “X”), Facebook, TikTok, and YouTube have achieved global proliferation with numerous active users, exerting considerable influence on societal discourse. These platforms enable users to freely discuss opinions and engage in mutual communication on topics ranging from pressing social issues, politics, and scientific concerns, to everyday information exchange. Through these discussions, users influence each other, form opinions, gain insights, and often change their perspectives.

Social media offers numerous benefits to human communication; however, some issues remain. For instance, users tend to follow like-minded accounts and unfollow those with different views, thereby creating personalized information environments. Social media platforms employ algorithmic curation systems that

filter content based on user behaviors, patterns, search histories, and interactions. Such personalization enhances the user experience, but has been cited as a factor in undesirable phenomena, such as opinion polarization and echo chambers. In computational social science, elucidating the mechanisms underlying opinion polarization and exploring countermeasures are recognized as urgent research objectives.

With the recent development of *large language models* (LLMs) in artificial intelligence (AI) technology, it has become easier to manipulate public opinion on social media during political elections using bot-automated accounts that mimic human behavior and amplify specific narratives. Because such AI-based bots have become more sophisticated, distinguishing their opinions from those of real people is difficult, which increases their influence. In addition, popular *hub users*, who have numerous followers and are known as *influencers*, play a significant role in online discourse and can be exploited for opinion manipulation through bribery or strategic tactics. **Although both hub users and bot-driven operations on social media have received considerable attention, research on hub users who shift from neutral to biased stances remains limited, despite increasing concerns about such manipulative behavior.**

Therefore, we investigate the effect of opinion manipulation by influencers with moderate opinions on user behavior, including opinion shifts among both hub and normal users. We propose a multiagent-based model of social media platforms on complex networks, characterized by scale-freeness, high clustering, and small-worldness. In this model, nodes represent users and directed edges represent follow relationships. The model incorporates key mechanisms, including *dynamic bounded confidence*, social rewards through reposts(likes), social reinforcement, and follow/unfollow actions, to reproduce phenomena such as echo chambers, vocal minorities, and opinion polarization. Experiments using this model, despite its simplicity as a combination of established mechanisms, reveal that manipulating moderate influencers provokes a strong, counter-intuitive backlash among users with opposing extreme opinions, increasing their activity and consequently reinforcing overall polarization. These findings highlight the unintended consequences of influencer-based opinion shaping on social media platforms and provide insights into the interplay between opinion polarization and the spiral of silence.

## 2 Related Work

Prior to social media, studies on social simulation explored the mechanisms of collective decision-making in public discourse and predicted public opinion dynamics. A common approach for exploring opinion dynamics is *multiagent-based simulation* (MABS) [13, 20], which simulates complex social dynamics based on simple models [10, 7] in which agents autonomously determine behaviors, including social interactions and information received from others. With the spread of social media, the number of studies analyzing real-world human behavioral data using MABS has increased [2, 23]. Social media platforms enable researchers to

collect large-scale data by tracking human activities, thereby allowing the empirical evaluation of social science theories [5, 18]. This research style, which combines computational modeling with data-driven analysis, is known as *computational social science*, a new paradigm for studying social activities [12].

Studies on social media have investigated opinion polarization and echo chambers. An echo chamber is a phenomenon in which users are isolated within a highly polarized cluster of communities, fostering misinformation and extreme opinions, leading to polarization while discouraging constructive dialogue-based diverse opinions, thereby undermining a democratic foundation. Numerous empirical studies have investigated echo chambers using real-world datasets from social network platforms [4, 18, 25, 2]. These studies focused on specific features of social media platforms, such as algorithmic recommendation systems, unfollow mechanisms, and design elements that lead to echo chambers when combined with human psychological features [17, 18].

Automated bot activities and influencer operations have drawn substantial academic and societal attention. Approximately 10% of all accounts on major social media platforms are estimated to be bots [22], prompting extensive research on bot detection and their influence on public discourse [8]. Goldstein et al. [8] examined the impact of generative large language models (LLMs) on campaign participants, behaviors, and content, emphasizing the emerging uncertainties surrounding future information operations. In parallel, prior studies have explored the mechanisms and impacts of opinion leaders, or influencers, through empirical analyses and simulation-based approaches [15, 16, 14]. Sanjay et al. [16] proposed an influence maximization framework using graph embeddings and graph neural networks, while Betts et al. [15] demonstrated that influencers’ effects on polarization depend on both their audience reach and their expressed opinions. However, limited research has examined how strategic opinion shifts—particularly among influencers who initially hold neutral positions—affect echo chambers and polarization dynamics. Such shifts, often driven by coercive strategies, bribery, or psychological manipulation by external actors, have been increasingly reported in real-world contexts. To address this gap, we develop an agent-based model that incorporates users’ behavioral characteristics to analyze the effects of influencer manipulation on the evolution of public opinion.

### 3 Opinion Dynamics Model

#### 3.1 Agent Model

We introduce an agent-based model for opinion dynamics on social media platforms. The social media environment is a directed graph  $G = (A, E)$ , where  $A$  denotes the set of  $n$  agents (users), and  $E$  denotes the set of directed edges representing the following relationships: for agents  $i, j \in A$  ( $i \neq j$ ), the directed edge from  $i$  to  $j$  indicates  $i$  follows  $j$ . We introduce a discrete time denoted by a nonnegative integer,  $t$ .

Table 1: Agent parameters and their value ranges

| Parameter                         | Symb.      | Range         | Init. value |
|-----------------------------------|------------|---------------|-------------|
| Opinion                           | $O_i$      | $[-1.0, 1.0]$ | Random      |
| Posting probability               | $\rho_i$   | $[0.1, 1.0]$  | 0.1         |
| Platform access probability       | $\sigma_i$ | $[0.1, 1.0]$  | 0.1         |
| Dynamic bounded confidence in BCM | $B_i$      | $[0.2, 1.0]$  | 1.0         |

Agent  $i$  is characterized by the parameters listed in Table 1, which express the individual attributes of opinions and posting behaviors. The opinion of  $i$  is represented as a one-dimensional vector  $O_i(t)$ , reflecting agreement or disagreement with an issue, ranging from  $-1$  (extremely negative) to  $1$  (extremely positive) at time  $t$ . If  $O_i(t)$  is approximately 0,  $i$  has moderate opinion. We refer to  $O_i(0)$  as the *intrinsic opinion* of  $i$  because this is the initial opinion unaffected by others. Parameter  $\sigma_i$  denotes the *platform access probability*;  $i$  may read articles in its feed at  $t$ ,  $F_i(t)$ , comprising recent  $N_F > 0$  unread articles posted by other agents, and then post based on their opinions with *posting probability*  $\rho_i$ . Otherwise,  $i$  neither reads nor posts their opinions. Note that these probabilities were constant in the previous research by Sasahara et al. [18].

We also employ the dynamic bounded confidence  $B_i \in [0.2, 1]$  in the *bounded confidence model* (BCM) [6], which reflects confirmation bias and homophily, and is widely used in social network analysis. The parameters listed in Table 1 are dynamic and evolve during interactions with neighboring agents according to the update rules described below. We introduce the following and unfollowed account lists  $L_i^f$  and  $L_i^u$  for agent  $i$ , expressing the agents (or accounts) that  $i$  follow and intentionally unfollow. Initially,  $L_i^f$  for  $\forall i \in A$  is set based on the initial content of  $E$  in  $G$  and  $L_i^u$  is an empty list. Note that  $E$  is modified at each time step according to these lists.

### 3.2 Modeling Social Media Behavior

To model user behavior on social media platforms, we integrate two mechanisms: (1) the *social adoption* of a platform and (2) *social actions*, namely, follow, unfollow, post, and repost.

**Social Adoption** Agent  $i \in A$  decides to use the platform with probability  $\sigma_i$  and updates the internal parameters using information from the surrounding environment every time  $t$ .

**Social Influence:** We adopt the Friedkin–Johnsen model (FJ) [7] to represent opinion updating under social influence; therefore, when  $i$  accesses the platform at time  $t$ , their opinion  $O_i(t)$  is updated as follows:

$$O_i(t) = \mu \cdot O_i(0) + (1 - \mu) \cdot \frac{\sum_{p_j \in F_i(t)} O(p_j)}{|F_i(t)|}, \quad (1)$$

where  $\mu$  denotes the agents' *stubbornness* toward the intrinsic opinion  $O_i(0)$ ;  $F_i(t)$  denotes the feed of  $i$ , the set of posts read by  $i$  at  $t$ ; and  $O(p_j) (\in [-1, 1])$  denotes the opinion of  $p_j$  posted by  $j$ . This formulation shows how agents update their opinions based on the average opinions of the posts in their feed, which are modulated by stubbornness.

**Social Reinforcement (SR):** Social media users commonly post opinions on platforms that are then shared with others. To model this, we introduce a reinforcement mechanism: if articles with opinions in  $F_i(t)$  similar to  $i$ 's opinion constitute the majority of  $F_i(t)$ ,  $i$  increases the posting probability  $\rho_i$  and platform access probability  $\sigma_i$ . Conversely,  $i$  decreases if they hold a minority opinion. Formally,

$$\text{If } C_i(t) \geq \varepsilon_s, \quad \sigma_i = \min(\sigma_i + \delta_\sigma, 1), \quad \rho_i = \min(\rho_i + \delta_\rho, 1) \quad (2)$$

$$\text{If } C_i(t) \leq (1 - \varepsilon_s), \quad \sigma_i = \max(\sigma_i - \delta_\sigma, 0), \quad \rho_i = \max(\rho_i - \delta_\rho, 0) \quad (3)$$

where  $C_i(t) = |K_i(t)|/|F_i(t)|$  and  $K_i(t) = \{p \in F_i(t) \mid |O_i(t) - O(p_j)| \leq \varepsilon_t\}$ . The parameters  $\varepsilon_t > 0$  and  $\varepsilon_s > 0$  are the *opinion tolerance* and *local opinion conformity*, respectively. The SR parameters  $\delta_\sigma \geq 0$  and  $\delta_\rho \geq 0$  denote the degree of the influence of (in)activation. When  $F_i(t) = \emptyset$ ,  $\sigma_i$  and  $\rho_i$  remain constant. In our experiment,  $\delta_\sigma = 0.01$  and  $\delta_\rho = 0.1$ .

We assume that agent  $i$  changes  $\rho_i$  based on the social acceptance of neighborhood members. Agent  $i$  increases the posting rate  $\rho_i$  by reward  $r_i$  based on reposts by agents reading articles posted by  $i$  as follows:

$$r_i = \alpha \cdot \ln(1 + N_i^l) \quad (4)$$

where  $\alpha$  is a tuning parameter and  $N_i^l$  is the number of reposts received after  $i$ 's previous access. As platforms propagate liked posts through reposts, we assume that reposts have equal numbers of likes. We use diminishing marginal utility [9] for the social reward; the utility from reposts is modeled using a logarithmic function of likes received.

**Reducing confidence:** We define post  $p_j$  from agent  $j$  as *comfortable* for agent  $i$  if  $|O_i(t) - O(p_j)| < B_i$ ; otherwise,  $p_j$  is *uncomfortable*. When  $i$  reads an uncomfortable post in  $F_i$ , confidence in  $i$  decreases as  $B_i = \max(0.2, B_i - \delta_b)$ , where  $1 \gg \delta_b > 0$ . This indicates that uncomfortable posts reduce confidence in public discussions on social media [1].

**Social Actions** Each time, agent  $i$  determines whether to access the platform with probability  $\sigma_i$ . If  $i$  does this, it performs the following actions:

**Follow:** If the feed of  $i$ ,  $F_i$ , contains at least one comfortable post from other agents that are not in the following list  $L_i^f$ ,  $i$  follows one of them (random selection) with the *following probability*  $P_+ > 0$  and adds the agent to  $L_i^f$ .

**Unfollow/Block:** If  $F_i$  contains at least one uncomfortable post from agents in the following list  $L_i^f$ , and  $L_i^f$  contains at least two agents,  $i$  unfollows one of them  $j \in A$  randomly with *unfollowing probability*  $P_-$  (adding  $j$  to  $L_i^u$ ,

removing  $j$  from  $L_i^f$ ). Otherwise, if at least one uncomfortable post exist from agents not in  $L_i^f$ ,  $i$  selects an uncomfortable post  $p_j$  and blocks agent  $j$  with probability  $P_-$  by adding  $j$  to the unfollowed account list  $L_i^u$ .

**Post:** Agent  $i$  posts an article expressing its opinion  $O_i(t)$  with probability  $\rho_i$ .

**Repost (Like):** After reading post  $p_j$  ( $j \in A$ ) in the feed, agent  $i$  reposts it with probability  $P_r$  if  $|O_i - O(p_j)| < B_i$ . Subsequently,  $p_j$  is added to the feeds of  $i$ 's followers, allowing  $p_j$  to diffuse through social media.

This agent-based model simplifies user behavior and interactions. However, we believe that this model can capture the fundamental dynamics of online discussions and provide clear insights into the complex outcomes of interactions.

### 3.3 Manipulation of Influencers

Our primary concern is the effect of influencers on opinion changes in a specific direction and analyze their ability to influence public opinion. We analyzed the manipulation by influencers and whether public opinion shifts toward the target opinion according to their intentions, regardless of neighbors' opinions, using the proposed model. Therefore, the targeted influencer  $i$  changes its opinion by

$$O_i(t+1) \leftarrow O_i(t) \pm \delta_o \quad (5)$$

without using the FJ model (Eq. (1)), where  $\delta_o \in [-1, 1]$  denotes the *opinion displacement parameter*. If  $O_i(t)$  exceeds the range  $[-1, 1]$ , it is set to the value nearest to  $\pm 1$ .

## 4 Experiments and Discussion

### 4.1 Experimental Setting

We used the *connecting nearest neighbor with random linkage* (CNNR) network [26] as the initial social media environment  $G = (A, E)$ . It is generated with the probability of creating an edge from a potential edge  $u = 0.3$  and random linkage probability  $r = 0.01$ . This network extends the *connecting nearest neighbor* model [24], which has three key properties of social networks: a scale-free structure, high clustering coefficient, and small world. This adds randomness to connections, creating more realistic networks. Table 3 lists the network metrics that characterize the initial network.

The agents' intrinsic opinions  $O_i(0)$  are assigned according to the normal distribution  $N(0, 0.6^2)$ . Because the random intrinsic opinions of influencers may cause imbalanced influences on opinion dynamics, we randomly reassign the fixed opinions of  $-1$ ,  $0$ , and  $1$  to the three agents with the largest degrees. The code for these experiments is available on GitHub.<sup>3</sup>

We set the initial posting and platform access probabilities of agent  $i$  to  $\rho_i = 0.1$  and  $\sigma_i = 0.1$ . The reposting probability was set to  $P_r = 0.4$  based

<sup>3</sup> <https://github.com/toooooomoya/backfire-of-manipulation>

Table 2: Experimental parameters

| Parameter                                     | Value |
|-----------------------------------------------|-------|
| Reposting probability of $P_r$                | 0.4   |
| (Un)following probabilities $P_+$ & $P_-$     | 0.1   |
| Stubbornness in social influence $\mu$        | 0.8   |
| Opinion tolerance $\varepsilon_t$             | 0.2   |
| Local opinion conformity $\varepsilon_s$      | 0.8   |
| SR param. for posting $\delta_p$              | 0.1   |
| SR param. for platform access $\delta_\sigma$ | 0.01  |
| Confidence reduction $\delta_b$               | 0.003 |
| Opinion displacement parameter $\delta_o$     | 0.01  |

Table 3: Initial network metrics

| Parameter              | Value |
|------------------------|-------|
| No. of agents, $ A $   | 1000  |
| No. of edges, $ E $    | 1477  |
| Average degree         | 1.434 |
| Average path length    | 3.093 |
| Diameter               | 11    |
| Power-law exponent     | 2.585 |
| Clustering coefficient | 0.110 |

on the average posting rates from a previous study [21]. We set the maximum feed size of all agents to  $|F_i| = N_F = 10$  [18] because users cannot read all the received articles. The parameter for SR rewards (Eq. (4)) was defined as  $\alpha = 0.1$ . The other social media behavioral parameters are summarized in Table 2. These values were determined by parameter scanning using real-world datasets [4].

## 4.2 Shift in Opinions and Polarization

**Shift in opinions** In the first experiment (Exp. 1), we investigated the variation of opinions in posts over time using the proposed model without manipulating influencers as a baseline scenario. The results are presented in Fig. 1, which represents the average of 100 experimental runs with different seeds (except for Figs. 1c and 1d, which are generated using one selected run). Figure 1a plots the number of all posts and the ratios of classified posts every 100 iterations. The posts are classified into five clusters by dividing the opinion range  $[-1, 1]$  into five subranges of 0.4 increments. The figure shows that the number of posts has constantly increased. Initially, posts with moderate opinions ( $-0.2 \leq O_i \leq 0.2$ ) were dominant and decreased over time, whereas posts with extreme opinions ( $-1 \leq O_i \leq -0.6$  and  $0.6 \leq O_i \leq 1$ ) were initially low, but increased gradually. Posts from users with extreme opinions eventually became dominant, accounting for approximately 60%. We believe that this trend is commonly observed in online discourse and is widely recognized as polarization.

In Fig. 1b, the variance in the apparent opinions which is calculated from the posts and the variance in the overall opinions of all agents are plotted. This figure shows that these variances differ, indicating that apparent opinions do not necessarily align with overall opinions. Before 1000 steps, the apparent opinion variance was lower than the overall opinion variance because posts with moderate opinions dominated. However, the apparent opinion variance subsequently exceeded the overall opinion variance. To the best of our knowledge, minimal emphasis has been placed on reproducing this phenomenon of *apparent polarization* using multiagent social simulations. The following subsections interpret

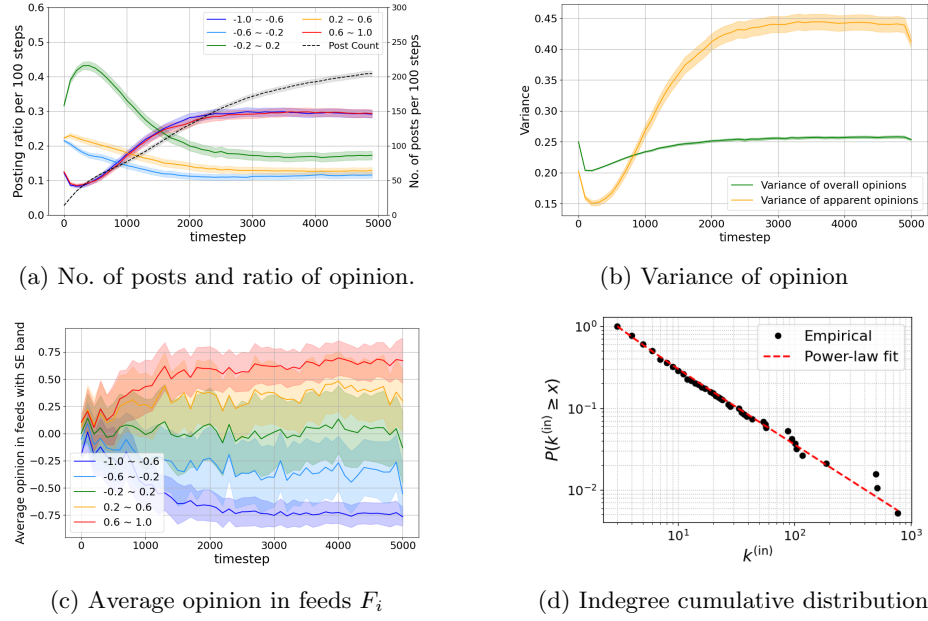


Fig. 1: Analysis of Opinion Shifts

these mechanisms in the proposed model and provide insights into the relationship between polarization and vocal minorities.

**Apparent polarization: vocal polarized minority** The degree of polarization, as shown in Fig. 1a, corresponds to the variance of apparent opinions in Fig. 1b, because both were calculated from post-opinions. These two indicators suggest the emergence of surface-level polarization. However, the actual variance of the overall opinions was smaller than that of the apparent opinions, indicating that the observed polarization was more pronounced than the underlying reality. Although agents with extreme opinions frequently post, regardless of the follower count, those with moderate opinions become less active, forming a silent majority, except for those with numerous followers. In our experiments, most agents had moderate opinions. This imbalance, in which a vocal minority dominates the discourse, triggers apparent polarization. This finding highlights the limitations of inferring public opinion solely from social media platforms.

**Trapped in echo chambers** We investigated the opinions expressed in posts read by each agent. We plotted the average values of opinions in feed  $F_i$  based on the subranges of opinion  $O_i(t)$  in Fig. 1c. The red curve indicates the average opinion of  $F_i$  for  $\forall i \in A$  s.t.  $O_i(t)$  ranges between  $-1.0$  and  $-0.6$ . The light-colored bands above and below the line represent standard errors. This graph



shows the results of one run; however, the same trend was observed for the other runs.

Figure 1c shows that the agents initially read diverse, unbiased opinions. The posts in an agent’s feed  $F_i$ , particularly in the feeds of agents with extreme opinions, gradually become biased, with most posts aligned with their opinions. This leads them to believe that others share their opinions, causing polarization. This indicates the formation of echo chambers. The information environment in the user feeds serves as a substitute for perceived public opinion. Users become trapped in echo chambers through SR mechanisms (see Section 3.2) and increasingly express their opinions through posts.

Figure 1d, plotting the in-degree cumulative distribution of agents at 5000 steps, shows the final environment  $G$  maintained scale-free property with power-law exponent 1.895. The other network metrics were an average path length 1.299, average in-degree 4.791, diameter 5.00, and clustering coefficient 0.203.

### 4.3 Affected polarization with manipulated influencers

In the second experiment (Exp. 2), we investigated the effect of manipulated influencers on post opinions by comparing the results with those without manipulation. In our scenario, the two highest in-degree influencers with moderate opinions (except the top three influencers because their opinions were fixed) were designated as manipulated influencers and forced to shift their opinions toward  $-1$  (depicted by blue curves in the figures) from the 1000th step to increase opinions close to  $-1$  (see Section 3.3). The results are shown in Fig. 2.

Exp. 2 reveals unintended results: an increase in posting activity by users with opinions near  $+1$ . Figure 2a shows the number of posts and opinion ratios in the divided subranges from one instance of the run in Exp. 1. In this instance, extreme opinions near  $-1.0$  (blue curve) were more prevalent than those near  $1.0$  (red curve). Figure 2a shows the same results when influencers manipulated posts. Comparing these figures, the manipulation led to an increase in opposing opinions after the 1000th step, whereas the intended opinion slightly diminished.

To analyze this phenomenon, we introduce the (*socially*) *reinforced agent*, who gained social confidence through the support of surrounding opinions, which is defined by Eq. (2) and was satisfied at least 80 times during the most recent 100 timesteps. We show the difference in the number of reinforced agents when the two influencers manipulate and do not in Fig. 2c, where positive values indicate an increase in the number of reinforced agents.

We believe that this phenomenon was caused by the backlash of agents with opposing opinions. The basic mechanism is as follows: (1) users with extreme opinions develop narrow-bounded confidence owing to exposure to conflicting posts; (2) moderate influencers performing manipulation have high indegrees, making their posts widely spread; (3) as they shift to intended opinions, followers with opposing opinions reduce bounded confidence and increase the likelihood of unfollowing them; and (4) losing moderate influencers from their follow list (followees) reduces the diversity of opinions appearing in users’ feeds, leading them to become trapped in highly closed echo chambers, thereby increasing the

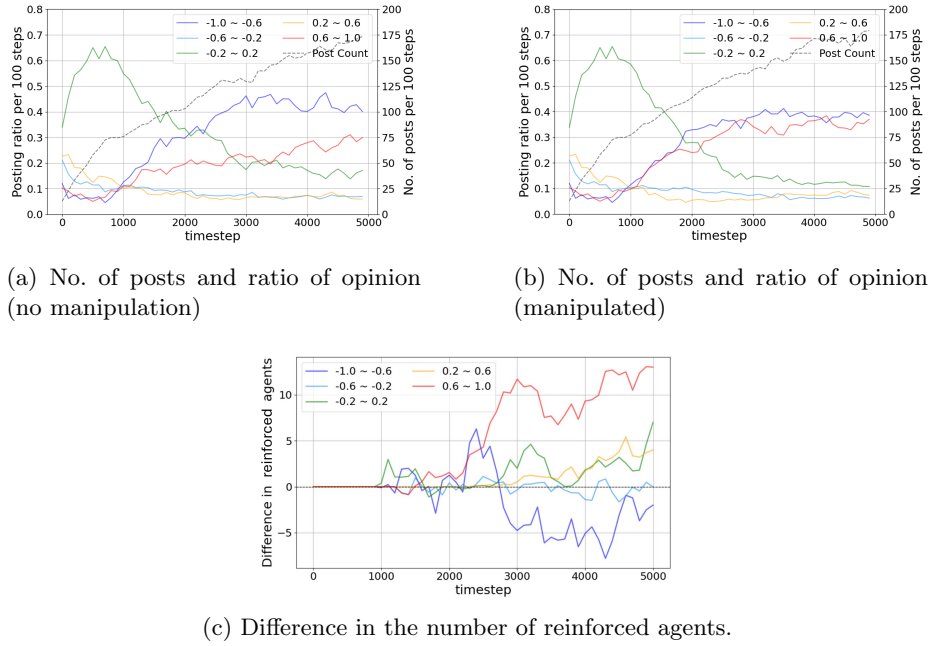


Fig. 2: Effects of Influencer Manipulation on Opinion Shifts and Polarization

number of reinforced agents, as shown in Fig. 2c. These results suggest that when well-connected and moderately connected users change their opinions, they can be perceived as ideological betrayals. This is a powerful trigger that further radicalizes users with opposing views.

## 5 Conclusion

We uncovered the mechanisms underlying the emergence of vocal minorities and their contribution to opinion polarization on social media using a multiagent model. Although the model combines well-established mechanisms, we observed counter-intuitive outcomes when influencers are manipulated.

First, we reproduced the polarization driven by vocal minorities, where agents with extreme opinions posted more frequently than moderate majority agents. This behavior stems from opinion-narrowed feeds, that is, the echo-chamber effect. Our analysis revealed that agents with extreme opinions were more likely to encounter conflicting opinions, thereby reducing their bounded confidence. This reduction accelerates selective exposure and posting behavior, reinforcing echo-chamber dynamics.

Second, we introduced influencers whose opinions were manipulated in a specific direction. Originally perceived as neutral and highly connected, these influencers delivered diverse opinions to their followers. When their opinions

shifted, opposing followers were more likely to unfollow them, trapping them in closed echo chambers and triggering a strong, counter-intuitive backlash among extreme users, thereby reinforcing overall polarization. These findings highlight the critical role of neutral, well-connected influencers and the structural mechanisms underlying unintended polarization in online social systems.

A limitation of this study is that the proposed agent based model is intentionally simplified to isolate the core mechanisms of polarization through opinion manipulation. In particular, it does not account for other forms of influence, such as algorithmic recommendations, and the simulations are restricted to networks generated by the CNNR algorithm. Future work should incorporate these complexities for a more comprehensive analysis.

## References

1. Baumeister, R.F., Bratslavsky, E., Finkenauer, C., Vohs, K.D.: Bad is stronger than good. *Review of General Psychology* **5**(4), 323–370 (2001). DOI 10.1037/1089-2680.5.4.323.
2. Buongiovanni, C., Candusso, R., Cerretini, G., Febbe, D., Morini, V., Rossetti, G.: Will you take the knee? Italian twitter echo chambers’ genesis during EURO 2020. In: *Complex Networks and Their Applications XI*, pp. 29–40. Springer International Publishing, Cham (2023)
3. Cinelli, M., Morales, G.D.F., Galeazzi, A., Quattrociocchi, W., Starnini, M.: Echo chambers on social media: A comparative analysis (2020). URL <https://arxiv.org/abs/2004.09603>
4. Conover, M., Ratkiewicz, J., Francisco, M., Goncalves, B., Menczer, F., Flammini, A.: Political polarization on twitter. *Proc. of the Int. AAAI Conference on Web and Social Media* **5**(1), 89–96 (2021). DOI 10.1609/icwsm.v5i1.14126.
5. Das, A., Gollapudi, S., Munagala, K.: Modeling opinion dynamics in social networks. In: *Proc. of the 7th ACM Int. Conf. on Web Search and Data Mining, WSDM ’14*, p. 403–412. ACM (2014). DOI 10.1145/2556195.2559896.
6. Deffuant, G., Neau, D., Amblard, F., Weisbuch, G.: Mixing beliefs among interacting agents. *Advances in Complex Systems* **03**(01n04), 87–98 (2000). DOI 10.1142/S0219525900000078.
7. Friedkin, N.E., Johnsen, E.C.: Social influence and opinions. *The Journal of Mathematical Sociology* **15**(3–4), 193–206 (1990). DOI 10.1080/0022250X.1990.9990069.
8. Goldstein, J.A., Sastry, G., Musser, M., DiResta, R., Gentzel, M., Sedova, K.: Generative language models and automated influence operations: Emerging threats and potential mitigations (2023). URL <https://arxiv.org/abs/2301.04246>
9. Gossen, H.H.: *The laws of human relations and the rules of human action derived therefrom*. MIT Press (1983). URL <https://cir.nii.ac.jp/crid/1971430859722206900>
10. Holley, R.A., Liggett, T.M.: Ergodic Theorems for Weakly Interacting Infinite Systems and the Voter Model. *The Annals of Probability* **3**(4), 643 – 663 (1975). DOI 10.1214/aop/1176996306.
11. Khanam, K.Z., Srivastava, G., Mago, V.: The homophily principle in social network analysis: A survey. *Multimedia Tools and Applications* **82**(6), 8811–8854 (2023). DOI 10.1007/s11042-021-11857-1.

12. Lazer, D., Pentland, A., Adamic, L., Aral, S., Barabási, A.L., Brewer, D., Christakis, N., Contractor, N., Fowler, J., Gutmann, M., Jebara, T., King, G., Macy, M., Roy, D., Alstyne, M.V.: Computational social science. *Science* **323**(5915), 721–723 (2009). DOI 10.1126/science.1167742.
13. Rainer, H., Krause, U.: Opinion dynamics and bounded confidence: Models, analysis and simulation. *Journal of Artificial Societies and Social Simulation* **5**(3) (2002)
14. Rothut, S., Schulze, H., Hohner, J., Rieger, D.: Ambassadors of ideology: A conceptualization and computational investigation of far-right influencers, their networking structures, and communication practices. *New Media & Society* **26**(12), 7120–7147 (2024). DOI 10.1177/14614448231164409.
15. Betts, John M. and Bliuc, Ana-Maria: The Effect of Influencers on Societal Polarization 2022 Winter Simulation Conference (WSC), 370–381 (2022). DOI 10.1109/WSC57314.2022.10015491.
16. Sanjay Kumar and Abhishek Mallik and Anavi Khetarpal and B.S. Panda: Influence maximization in social networks using graph embedding and graph neural network *Information Sciences* **607**(8), 1617–1636 (2022). DOI 10.1016/j.ins.2022.06.075.
17. Santos, F.P., Lelkes, Y., Levin, S.A.: Link recommendation algorithms and dynamics of polarization in online social networks. *Proc. of the National Academy of Sciences* **118**(50), e2102141,118 (2021). DOI 10.1073/pnas.2102141118.
18. Sasahara, K., Chen, W., Peng, H., Ciampaglia, G.L., Flammini, A., Menczer, F.: Social influence and unfollowing accelerate the emergence of echo chambers. *Journal of Computational Social Science* **4**(1), 381–402 (2021). DOI 10.1007/s42001-020-00084-7.
19. Sherman, L.E., Hernandez, L.M., Greenfield, P.M., Dapretto, M.: What the brain ‘likes’: neural correlates of providing feedback on social media. *Social Cognitive and Affective Neuroscience* **13**(7), 699–707 (2018). DOI 10.1093/scan/nsy051.
20. Sood, V., Antal, T., Redner, S.: Voter models on heterogeneous networks. *Phys. Rev. E* **77**, 041,121 (2008). DOI 10.1103/PhysRevE.77.041121.
21. Stieglitz, S., and, L.D.X.: Emotions and information diffusion in social media—sentiment of microblogs and sharing behavior. *Journal of Management Information Systems* **29**(4), 217–248 (2013). DOI 10.2753/MIS0742-1222290408.
22. Subrahmanian, V., Azaria, A., Durst, S., Kagan, V., Galstyan, A., Lerman, K., Zhu, L., Ferrara, E., Flammini, A., Menczer, F.: The darpa twitter bot challenge. *Computer* **49**(6), 38–46 (2016). DOI 10.1109/MC.2016.183
23. Toriumi, F., Sakaki, T., Kobayashi, T., Yoshida, M.: Anti-vaccine rabbit hole leads to political representation: the case of twitter in japan. *Journal of Computational Social Science* **7**(1), 405–423 (2024). DOI 10.1007/s42001-023-00241-8.
24. Vázquez, A.: Growing network with local rules: Preferential attachment, clustering hierarchy, and degree correlations. *Phys. Rev. E* **67**, 056,104 (2003). DOI 10.1103/PhysRevE.67.056104.
25. Waller, I., Anderson, A.: Quantifying social organization and political polarization in online platforms. *Nature* **600**(7888), 264–268 (2021). DOI 10.1038/s41586-021-04167-x.
26. Yuta, K., Ono, N., Fujiwara, Y.: A gap in the community-size distribution of a large-scale social networking site (2007). URL <https://arxiv.org/abs/physics/0701168>